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Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

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Chapter 1

Introduction

In recent years, thermal infrared (TIR) remote sensing from unmanned aerial vehicles (UAVs) and other airborne platforms has become an established approach for mapping thermal heterogeneity in river systems (Casas-Mulet et al., 2020; Dugdale, 2016; Kuhn et al., 2021; Torgersen et al., 2001). Mapping this heterogeneity is ecologically important: the cold-water patches it reveals can act as refugia that are essential for the survival of thermally sensitive fish species, particularly as river temperatures in Switzerland are expected to rise due to climate change. Despite this importance, there is currently no comprehensive information about the presence, scale, and location of such cold-water patches in Swiss water bodies (Tonolla and Antonetti, 2021, 2026).

As part of a cantonal project aiming to quantify the temperature heterogeneity and detect cold-water patches in the Emme River and its tributaries, a helicopter-based survey was previously conducted to collect TIR imagery covering over a 100 km river transect. These TIR images were processed using a newly developed algorithm for the automatic detection of cold-water patches, extracting longitudinal and transversal temperature gradients alongside the detected cold-water spots. Detailed information regarding the full workflow can be found in (Tonolla and Antonetti, 2021, 2026); the present thesis builds directly on this existing algorithm and the resulting dataset.

The CWP detection algorithm relies on externally configured parameters, and sensitivity analyses performed in (Tonolla and Antonetti, 2026) indicate that the detection results can vary significantly based on the chosen parameter values, with the step length and the temperature delta parameters having a particularly strong influence. The algorithm divides the river reach into fixed-length segments and computes a separate local reference temperature for each segment; visual inspection of existing detection results revealed that this can produce abrupt cut-off artefacts, where a cold-water patch spanning two adjacent segments is compared against two different reference temperatures and truncated at the segment boundary, even though the underlying thermal anomaly continues beyond it. Because cold-water tributaries tend to produce elongated thermal anomalies extending substantially along the river channel, such patches are disproportionately likely to span a segment boundary, suggesting that step length sensitivity may be elevated specifically for tributary-caused CWPs. A systematic and more extensive sensitivity analysis of the algorithm is therefore needed to test this assumption, to better characterise the resulting variance more broadly, and to inform potential algorithmic improvements. Performing such an analysis requires executing the algorithm across a large number of parameter combinations. The existing algorithm was found to require between 1.25 and 2 hours per

run on the available hardware, rendering a full sensitivity analysis infeasible without a faster implementation. A speed-optimized reimplementation is therefore a prerequisite for the sensitivity analysis itself.

Research Questions

Based on the above, this thesis pursues three objectives: first, to investigate whether the existing CWP detection algorithm can be reimplemented in a computationally more efficient form while producing equivalent outputs; second, to systematically investigate the algorithm's sensitivity to its parameters, with particular focus on how the presence of tributary inflows interacts with this sensitivity; and third, to derive adjustments to the detection algorithm that could reduce this parameter sensitivity while maintaining ecologically plausible detection results. These objectives are formalised in the following three research questions:

Research question 1 Can the existing CWP detection algorithm be reimplemented in a computationally more efficient form, while producing outputs that are equivalent to those of the original implementation?

Research question 2 How does the step length parameter contribute to the variance of the detected CWP area for tributary-caused CWPs, compared to non-tributary-caused CWPs?

H_0 The sensitivity of the detected CWP area to the step length parameter does not differ between tributary-caused and non-tributary-caused CWPs.

H_1 The sensitivity of the detected CWP area to the step length parameter is significantly higher for tributary-caused CWPs than for non-tributary-caused CWPs.

Research question 3 Building on the findings from research questions 1 and 2: what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results?

Chapter 2

Theoretical Background

2.1 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is “The study of how uncertainty in the output of a model [...] can be apportioned to different sources of uncertainty in the model input” (Saltelli, 2004). This definition rests on a broad notion of what counts as a model input: in sensitivity analysis, everything that causes variation in the output of the model is classified as an input (Saltelli et al., 2007). Providing an example: if a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \dots, X_n$ considered as model inputs, but also the parameters $\theta_1, \dots, \theta_n$ that correspond to the predictors. So the model:

$$Y = f(X_1, X_2, \theta_1, \theta_2) \tag{1-1}$$

would have four model inputs from the view of sensitivity analysis.

Sensitivity analysis provides a range of benefits. Firstly it allows the modeller to apply a strategy called “Factor Fixing”: if the model output Y is found to be independent of one or multiple inputs X_i , then these can be set to fixed constants, making sensitivity analysis a tool for model simplification (Saltelli et al., 2007). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical model components (Ligmann-Zielinska, 2013).

2.2 The Morris Method

The Morris method belongs to the group of screening methods in sensitivity analysis. Screening methods require only a small number of model evaluations, making them computationally cheap and therefore well suited to models that are expensive to run (Campolongo et al., 2007; Saltelli et al., 2007). The resulting sensitivity indices cannot be interpreted quantitatively, but they do allow for a qualitative, ordinal ranking of which model inputs are more important than others (Campolongo et al., 2011). As a global method, Morris explores the entire input space by varying all input parameters simultaneously across their plausible ranges, meaning that sensitivity scores reflect average behaviour across the full parameter space rather than being tied to a single point (Saltelli and Annoni, 2010). The following explanation of the Morris method closely follows (Saltelli et al., 2007) and is

explained more explicitly in chapter 3 of their book.

2.2.1 Elementary Effects and the Sampling Grid

At the core of the Morris method sits a k -dimensional model Y with inputs $X_1, \dots, X_k$. The model inputs are defined in the k -dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$. It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods. In the Morris method, this input space is discretized into a grid Ω , which is used as a basis for the computation of so-called elementary effects.

An elementary effect of the i th input parameter X_i is defined as:

$$EE_i = \frac{Y(X_1, X_2, \dots, X_{i-1}, X_i + \Delta, \dots, X_k) - Y(X_1, X_2, \dots, X_k)}{\Delta} \quad (2-2)$$

Where Δ represents some step along the grid Ω . Figure 3-3, taken from (Saltelli et al., 2007), shows an example of such a grid for a two-parameter, four-level case.

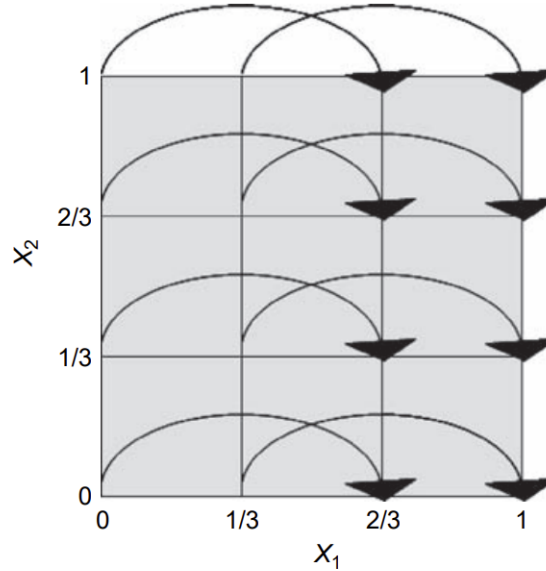


Fig. 2-1: An example grid Ω . The grid features two parameters X_1, X_2 , meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$ and indicated by the arrows.

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i , in other words $\mathcal{F}_i \sim EE_i$ (Saltelli et al., 2007).

2.2.2 Trajectory-Based Sampling

To populate $\mathcal{F}_i$ with multiple samples of EE_i , the Morris method uses an efficient trajectory-based sampling design that obtains as many elementary effects EE_i from as few model evaluations as possible. The exact mathematical formulation of how trajectories are constructed is beyond the scope of this thesis; a visual understanding is sufficient. Figure 2-2,

taken from (Saltelli et al., 2007), shows an example trajectory through a three-dimensional sampling grid.

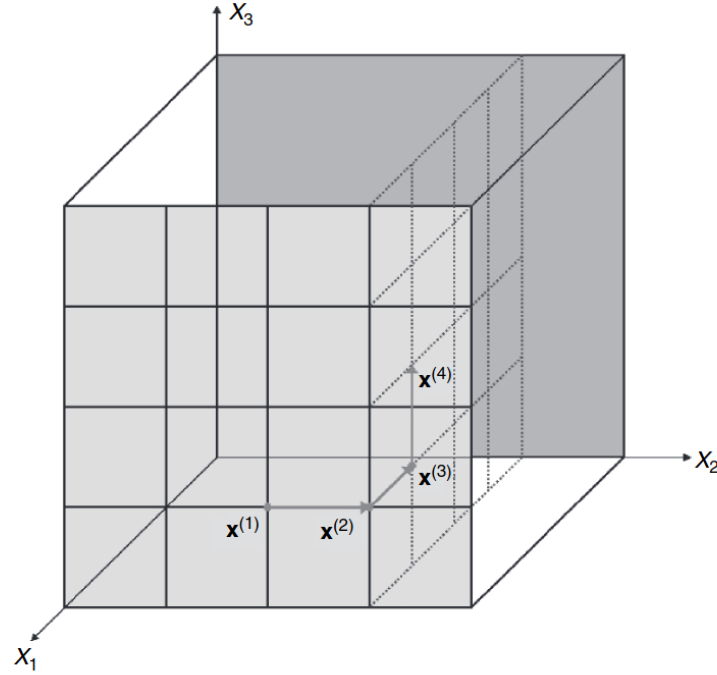


Fig. 2-2: An example Morris trajectory through a three-dimensional sampling grid ($k = 3$, $p = 6$). The trajectory consists of $k + 1 = 4$ points.

From this figure it becomes visible that a trajectory consists of $k+1$ points $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(k+1)}$, where k is the dimension of the sampling grid. Each point differs from its predecessor by a step of Δ in exactly one grid direction, so that consecutive points differ in only one model input X_i . The trajectory is constructed such that each dimension of the grid is stepped along exactly once. Each trajectory yields exactly k elementary effects EE_i – one for each parameter spanning the sampling grid – and requires $k + 1$ model evaluations.

To build up $\mathcal{F}_i$ for every factor, r such trajectories are generated. Since each trajectory contributes exactly one elementary effect per factor, each $\mathcal{F}_i$ ends up containing r samples. The total number of model evaluations required therefore is:

$$N = r \cdot (k + 1) \quad (2-3)$$

(Saltelli et al., 2007).

2.2.3 Choosing Δ and r

The choice of r and Δ is not independent of p . When p is chosen to be even, the choice $\Delta = \frac{p/2}{p-1}$ is ideal, as it has the additional property of guaranteeing equal-probability sampling from the grid, meaning that all grid levels are sampled with equal probability across the r trajectories (Saltelli et al., 2007). There is no clear rule regarding the number of trajectories r to pick. Generally, as the number of levels p in the grid increases, the number of trajectories r also needs to increase. Otherwise, increasing p serves little purpose, as the grid will not be sufficiently explored to feature all levels if the number of trajectories

is too small (Saltelli et al., 2007). Generally, the number of trajectories is usually chosen somewhere between 10–50 (Campolongo et al., 2007).

2.2.4 Sensitivity Measures: μ , σ , and μ^*

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed, specifically the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are of interest. These two statistical measures together already allow for reasonable insight into the influence of the i th parameter X_i , which corresponds to the i th elementary effect EE_i , onto the model:

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i . This ultimately means that changes in X_i generally affect the model output Y considerably, and that X_i therefore receives a high sensitivity score.

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates that the computed elementary effects EE_i generally differed depending on their location in the grid. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which of the two effects is responsible for the high standard deviation – or whether both contribute simultaneously – cannot be determined, which is a shortcoming of the Morris method (Saltelli et al., 2007). Figure 2-3 provides a visual example of possible $\mathcal{F}$ along with the associated μ and σ values and their corresponding interpretation.

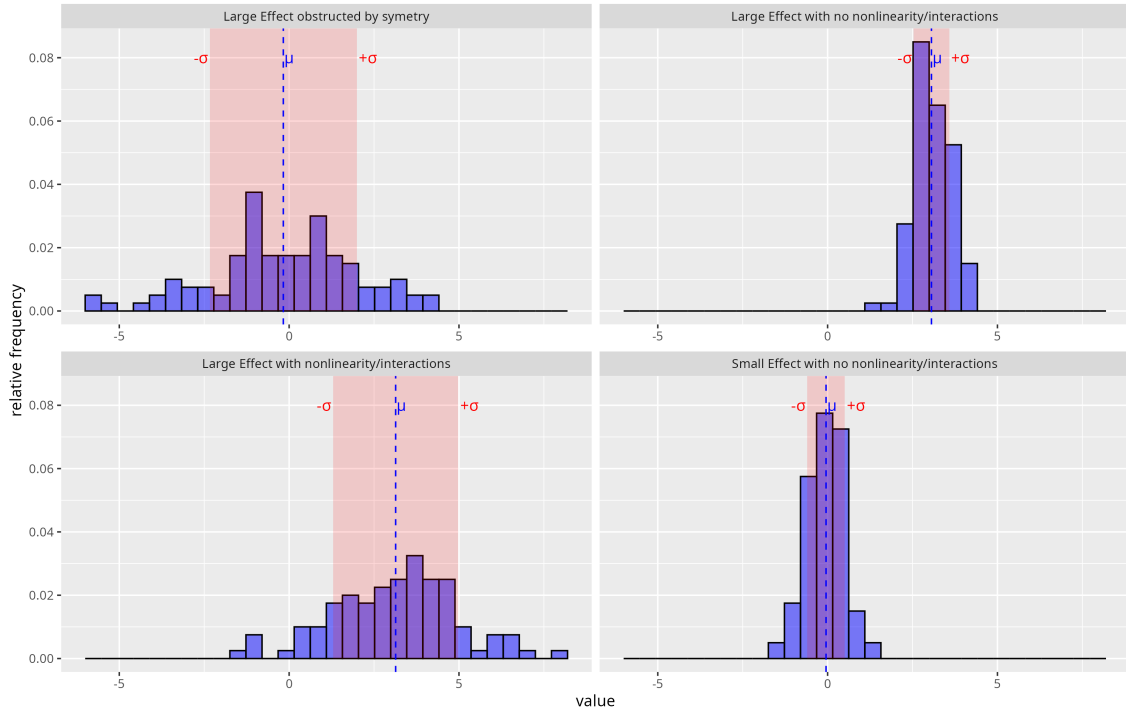


Fig. 2-3: Four example distributions $\mathcal{F}_i$ of a parameter X_i , illustrating how μ and σ jointly characterise parameter influence. *Top left:* a large effect obscured by symmetry. *Top right:* a large effect with no nonlinearity or interactions. *Bottom left:* a large effect with nonlinearity or interactions. *Bottom right:* a small effect with no nonlinearity or interactions.

Situations can occur where $\mathcal{F}_i$ contains many extreme values, yet the distribution is symmetric around zero. One such example is visualized in the upper-left plot of Figure 2-3. In such cases, computing the mean value μ_i yields a value close to zero, leading to the assumption that the parameter X_i is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in magnitude but cancel each other out. This phenomenon can be detected by always taking σ_i into consideration, since in this scenario σ_i is large. It is, however, more common to additionally sample a distribution $\mathcal{G}_i$, which captures the magnitude of the elementary effects $|EE_i|$. We then compute the mean of this distribution, μ_i^* , which can be interpreted like μ_i . Since it only captures magnitude, this distribution however never features negative values, and cancellation effects can therefore not occur (Saltelli et al., 2007). Figure 2-4 provides a visual example of this mechanism.

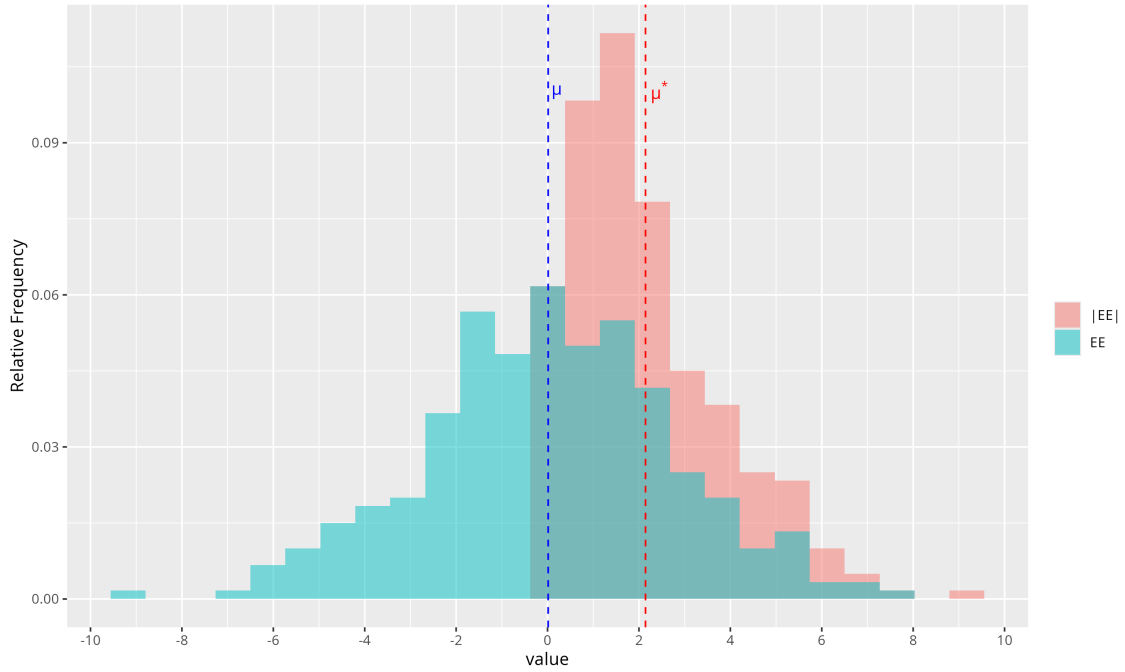


Fig. 2-4: The difference of μ and μ^* visualized with their underlying distribution $\mathcal{F}_i$ and $\mathcal{G}_i$. While the μ lies at around zero, μ^* significantly deviates from zero.

2.3 Application to Spatial Models

The Morris method, like most global sensitivity analysis methods, is defined for models with a scalar output: a single value Y is required for each model evaluation (Saltelli et al., 2007). The CWP detection algorithm, however, produces a spatially distributed output. Applying the Morris method to such a model therefore requires a spatial aggregation step to aggregate the spatial output to a scalar value, e.g. using a statistical aggregation function such as the mean (Lilburne and Tarantola, 2009).

2.4 Coldwater Patches

Cold water patches (CWPs) are areas in rivers and streams where the water temperature is measurably colder than that of the main channel. The temperature difference used to

define a CWP varies across the literature but is typically set between 0.5 and 3 degrees Celsius below the main channel temperature ([Casas-Mulet et al., 2020](#); [Wawrzyniak et al., 2016](#)). Cold water patches are not automatically cold water refugia. Cold water refugia are cold water patches that are actively used by cold-water organisms for thermal relief ([Mejia et al., 2023](#)). The ecological importance of cold water refugia is expected to increase as river temperatures rise due to climate change ([Kurylyk et al., 2015](#)). CWPs can arise from a range of physical mechanisms. A common cause is the inflow of a colder tributary-. Other generating mechanisms include groundwater upwelling, lateral seeps or hyporheic exchange ([Mejia et al., 2023](#)).

Chapter 3

Methods

This chapter describes the operationalisation of the three research questions into a set of methodological steps, which are then described in detail. The first section addresses research question 1 by describing the reimplementation of the CWP detection algorithm and the comparison of its outputs and runtime against the original. The second section addresses research question 2 by describing the Morris sensitivity analysis, including the selection and classification of CWPs, the computation of sensitivity indices, and the subsequent visualisation and statistical modelling. The third section addresses research question 3 by describing the algorithm modifications derived from the sensitivity analysis findings. A description of the data and study area is provided first.

Chapter 4

Methods

This chapter describes the operationalisation of the three research questions into a set of methodological steps, which are then described in detail. The first section addresses research question 1 by describing the reimplementation of the CWP detection algorithm and the comparison of its outputs and runtime against the original. The second section addresses research question 2 by describing the Morris sensitivity analysis, including the selection and classification of CWPs, the computation of sensitivity indices, and the subsequent visualisation and statistical modelling. The third section addresses research question 3 by describing the algorithm modifications derived from the sensitivity analysis findings. A description of the data and study area is provided first.

4.1 Data and Study Area

The Emme river originates in an extensive moorland between the Hohgant and Augstmattthorn massifs and flows through the Emmental region before joining the Aare near Zuchwil/Luterbach. The catchment covers approximately 974 km² with a mean elevation of 845 m a.s.l., and land cover is dominated by agricultural land (52.1%) and forested areas (37.9%). During summer thunderstorms, the river can rise rapidly within minutes and transport large quantities of driftwood; as a consequence, the lower reaches have been heavily engineered ([Tonolla and Antonetti, 2026](#)).

[Tonolla and Antonetti \(2026\)](#) acquired high-resolution TIR and RGB orthophotos along a 69 km transect of the Emme main channel, spanning from Kemmeribodenbad BE to Biberist SO, as well as imagery of three tributary rivers: the Ilfis, Grüene, and Urtene. The TIR imagery was collected from a helicopter platform using a FLIR SC655 thermal infrared camera. In this thesis, only the TIR and RGB orthophotos of the 69 km main channel section are used. The TIR orthophotos serve as the primary input to the CWP detection algorithms, while the RGB orthophotos are used in a downsampled state as background images for selected visualisations and were not used for any further analytical purpose. The 69 km river transect is covered by three partially overlapping single-band TIR orthophotos. Figure 1-1 provides an overview of the study area, showing the settlements demarcating each segment, the three orthophoto extents, and the three imaged tributaries. Table 1-1 summarises their technical specifications.

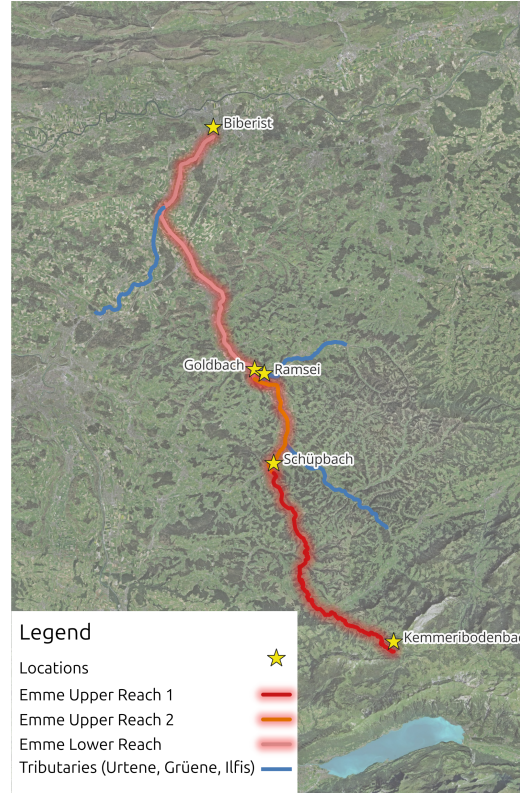


Fig. 1-1: Overview of the study area along the Emme river. The map shows the settlements bounding each of the three TIR orthophoto segments (Emme Upper Reach 1, Emme Upper Reach 2, Emme Lower Reach), as well as the three tributary rivers imaged during the survey (Ilfis, Grüene, and Urtene), shown here as a single combined category.

Tab. 1-1: Technical specifications of the three TIR orthophotos covering the Emme main channel. All rasters are in the Swiss coordinate reference system CH1903+/LV95 (EPSG:2056) and stored as single-band Float32 GeoTIFF files.

	Emme Reach 1	Upper Reach 2	Emme Reach	Upper Reach	Emme Reach	Lower Reach
Extent	Kemmeribodenbad – Schüpbach		Schüpbach – Goldbach		Ramsei – Biberist	
Total pixels	4.75×10^9		8.45×10^8		6.57×10^9	
Resolution (m)	0.207×0.207		0.204×0.204		0.207×0.207	
Min temp. (°C)	7.43		10.08		12.50	
Max temp. (°C)	33.67		30.94		34.10	
Mean temp. (°C)	20.85		18.84		20.90	
Std dev. (°C)	1.05		1.24		1.32	

4.2 Addressing Research Question 1: Algorithm Reimplementation

Performing the Morris sensitivity analysis requires $r \times (k + 1)$ model evaluations (Equation 2-3); for $k = 3$ parameters and $r = 64$ trajectories this amounts to 256 executions.

Given that the existing CWP algorithm requires between 1.25 and 2 hours per run, parallel execution on ZHAW's High Performance Cluster (HPC) is necessary to complete the experiment within a feasible timespan. To ensure that optimisation gains carry over to the HPC infrastructure, all benchmarking was performed directly on the cluster. In the remainder of this thesis, the original implementation is referred to as v1 and the speed-optimized reimplementation as v2.

4.2.1 Reimplementation of the CWP Algorithm based on GDAL and System Calls

The reworked algorithm preserves the computational logic of the original implementation but differs fundamentally in how each step is mechanistically executed. Where the original algorithm processes the raster slab-by-slab in memory using R loops and `terra/sf` functions, the reworked algorithm replaces these with calls to the GDAL command line interface, operating directly on raster and vector files on disk rather than holding intermediate results in memory. Table 2-2 gives a high-level overview of the reimplemented algorithm; the tool or function used in each step is listed in the right-hand column. The remainder of this section motivates the key design decisions underlying this choice of implementation.

Tab. 2-2: High-level overview of the reworked CWP detection algorithm, including the primary tool or function used for each step.

Step	Description	Tool / Function
1	Divide the river reach into n centerline segments; construct a wide zone polygon and a narrow reference-strip polygon for each segment.	<code>sf::st_buffer</code>
2	Rasterize the zone polygons to a raster matching the TIR raster resolution, using the segment id as burn-in value.	<code>gdal vector rasterize</code>
3	Round the TIR raster using a fixed rounding factor.	<code>gdal raster calc</code>
4	Extract the median temperature per segment from the rounded TIR raster within each reference strip; store in a look-up table keyed by segment id.	<code>exactextractr::exact_extract</code>
5	Reclassify the zone raster, replacing each segment id with its reference temperature from the look-up table, yielding the reference raster.	<code>gdal raster reclassify</code>
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.	<code>gdal raster calc</code>
7	Polygonize the binary CWP raster into patch polygons.	<code>terra::as.polygons</code>
8	Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.	<code>exactextractr::exact_extract</code>

Performance. The decision to heavily rely on GDAL was made as `terra` operations are sometimes not performant, and GDAL offers an often faster set of geocomputational tools, which can be called from R using `system()` or `sf::gdal_utils()` (Lovelace et al., 2019). Empirical testing in a prior coursework exercise also showed a significant speedup of GDAL-based raster operations compared to `terra` across a range of use cases (Dürig, 2026); this coursework was not peer-reviewed and is cited here only as a supporting empirical observation.

Memory footprint and HPC suitability. Operating directly on disk rather than in memory keeps the memory footprint small, which is important in an HPC setting where available memory can become a limiting factor under parallel execution. The drawback is that intermediate rasters can reach up to 25 GB in size; this was addressed by applying `COMPRESS=DEFLATE` with `PREDICTOR=2` when writing GeoTIFF files (GDAL/OGR

[contributors, 2026](#)), which substantially reduces disk space consumption with negligible computational overhead.

Raster value extraction. `terra::extract()` was replaced with `exact_extract()` from the `exactextractr` package (steps 4 and 8 of Table 2-2), which extracts aggregated raster values considerably faster for large rasters ([Mieno, 2022](#)). The two functions differ in how partially covered cells are handled: `exact_extract()` weights cells by their coverage fraction ([ISciences, LLC, 2026](#)), whereas `terra::extract()` includes a cell only if its centre point falls within the polygon ([Hijmans et al., 2024](#)) – the behaviour used in v1. To approximate this, only cells with a coverage fraction of at least 50% were included in `exact_extract()`; this approximation is not exactly equivalent to `terra`’s centre-point rule.

4.2.2 Output Equivalence and Runtime Comparison

The resulting reimplementaion was tested for equivalence in output by comparing the CWP polygons produced by v2 against those produced by v1 when deployed with the exact same set of parameters. Spatial agreement was assessed using the Jaccard similarity index:

$$J(A, B) = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)} \quad (2-1)$$

where A and B are the CWP polygon sets produced by v1 and v2 respectively. $J = 1$ indicates perfect agreement and $J = 0$ indicates no overlap. In addition, visual inspection in QGIS was carried out to identify any systematic spatial patterns in the disagreements. Table 2-3 shows the parameter combinations used in testing.

Runtime was compared across three segment lengths (400, 800, and 1200 m), with all other parameters held constant (zone halfwidth = 60 m, buffer = 3 px, $\Delta T = 1.0^\circ\text{C}$, minimum patch area = 2 m², rounding = 0.1 °C, 8-connectivity enabled).

Tab. 2-3: Parameter combinations used for the output equivalence comparison between v1 and v2.

Parameter	Test 1	Test 2	Test 3
segment_length_m	200	800	1200
zone_halfwidth_m	60	60	60
buffer_px	4	3	5
delta_C	0.5	1.0	1.5
min_patch_area_m2	2	3	2
round_to	0.1	0.1	0.1
connect_diagonals	TRUE	TRUE	TRUE

4.3 Addressing Research Question 2: Sensitivity Analysis

With the algorithm optimized, sensitivity analysis can now be carried out to determine the sensitivity of the CWP detection algorithm with respect to the parameters buffer pixel

count, step length, and temperature delta. The first step focuses on determining CWPs that are susceptible to the step length parameter and classifying them into tributary-caused and non-tributary-caused CWPs. The second step derives Morris sensitivity indices for these CWPs with respect to their area. The third step displays the results visually and fits a statistical model relating step-length sensitivity to CWP area and temperature delta.

The inclusion of buffer pixel count and temperature delta alongside step length also serves a secondary, validating purpose. [Tonolla and Antonetti \(2026\)](#) found that the buffer pixel count has little influence on CWP area, while the temperature delta parameter was among the most influential. Replicating both findings within the present methodology would lend additional credibility to both approaches.

4.3.1 Determining and Classifying Step Length Sensitive CWP

Especially of interest is to observe the difference in sensitivity towards the step length parameter between tributary-caused and non-tributary-caused CWP. For this reason, a pre-selection method is deployed to guarantee that the studied CWP are sensitive towards the step length parameter. A Morris design with $r = 44$ and $k = 2$ is first set up for the buffer pixel count and step length parameters. Then the speed-optimized (v2) CWP algorithm is deployed on the resulting parameter tuples. Lastly, the results are aggregated to a fuzzy representation of the CWP over all runs, to then select CWP which are variable with respect to the CWP algorithm and to classify them into tributary-caused and non-tributary-caused CWP. Figure 3-2 provides a visual overview of these steps.

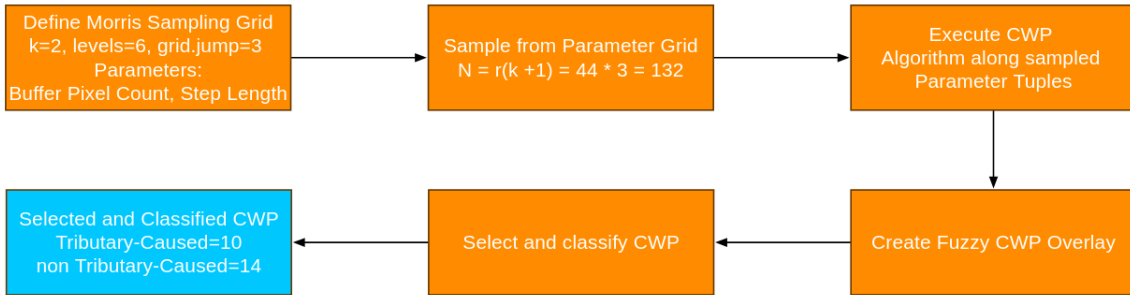


Fig. 3-2: Visual overview of the steps involved in determining and classifying step-length-sensitive CWP. The result is a set of classified CWPs.

Setting up First Morris Experiment

A two-dimensional Morris sampling grid was set up spanning buffer pixel count and step length, with 6 levels per parameter (step length: 400–1200 m; buffer: 2–7 px), using the R package `sensitivity` to generate the sampling design. In this package, the step size is not set directly as a continuous value Δ , but rather through the `grid.jump` argument, an integer specifying the number of grid levels to be jumped when computing each elementary effect. A `grid.jump` of 3 was chosen because it corresponds to the recommended step size (see 2.2.3), which implements equal-probability sampling of the elementary effects, given by:

$$\Delta = \frac{p}{2(p-1)} = \frac{6}{2 \times 5} = \frac{3}{5} \quad (3-2)$$

Since the spacing between adjacent levels of the 6-level grid in the normalized unit space is $1/(p - 1) = 1/5$, jumping 3 grid levels yields exactly $3 \times 1/5 = 3/5$, matching the recommended Δ . Figure 3-3 shows the grid with the jump distance drawn in. A total of 44 trajectories were sampled ($r = 44$), yielding $44 \times (2 + 1) = 132$ parameter tuples, each holding a unique combination of buffer pixel count and step length.

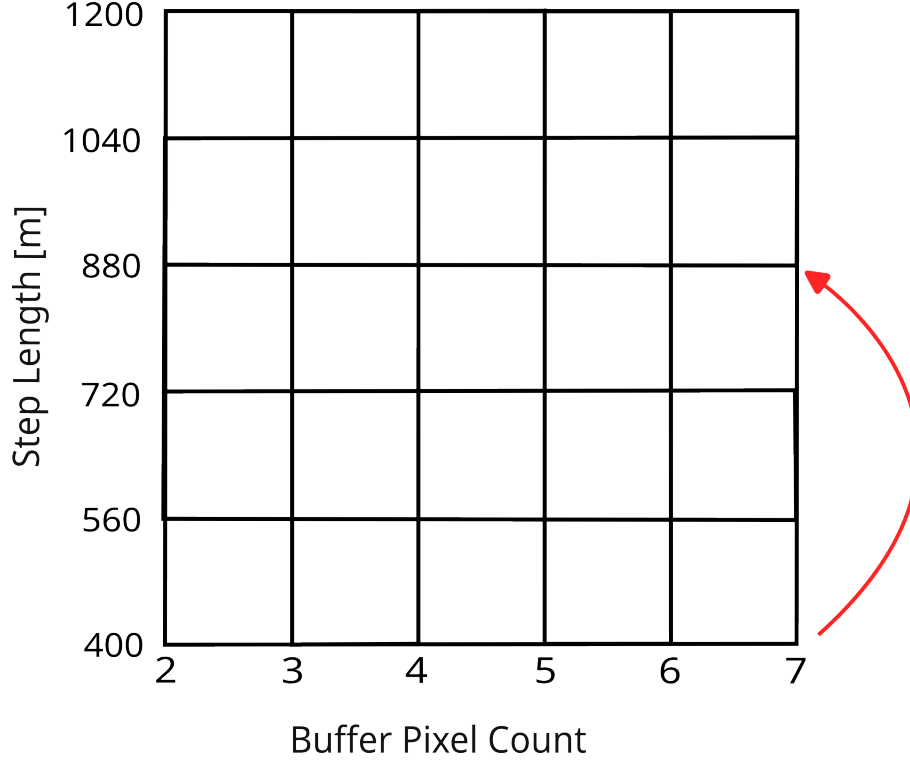


Fig. 3-3: Visual depiction of the Morris sampling grid, spanning step length and buffer pixel count. The arrow shows the Δ chosen for the Morris experiment, corresponding to a jump of 3 grid cells.

The temperature delta parameter was held constant ($k = 2$) because varying it would introduce variability in the detected extent of essentially all CWPs – making it impossible to distinguish CWPs that are sensitive to step length from those that merely respond to a changing detection threshold. Holding it constant ensures that any variability in the fuzzy overlay can be attributed to step length and buffer pixel count, and in particular to step length, as the buffer pixel count is generally non-influential. Further, a Morris design was used rather than a simple grid sweep to ensure construct validity of the selection criterion: any variability visible in the fuzzy overlay is, by construction, of the kind that the subsequent Morris sensitivity analysis is designed to detect. Had a different sampling scheme been used, visually variable CWPs might not register as sensitive under the Morris indices computed in the second experiment.

Model Evaluation along Parameter Combinations

The CWP algorithm was evaluated across all sampled parameter tuples in parallel using GNU Parallel (Tange, 2024), with the remaining parameters held constant as listed in Table 3-4. Each worker process fetched a parameter tuple, created a uniquely named working directory, and wrote all results there for traceability. Figure 3-4 visualizes the

workflow.

Tab. 3-4: Parameter values held constant during the first Morris experiment, while step length and buffer pixel count were varied.

Parameter	Value
zone_halfwidth_m	60 m
delta_C	1.0 °C
min_patch_area_m2	2 m ²
connect_diagonals	TRUE
round_to	0.1

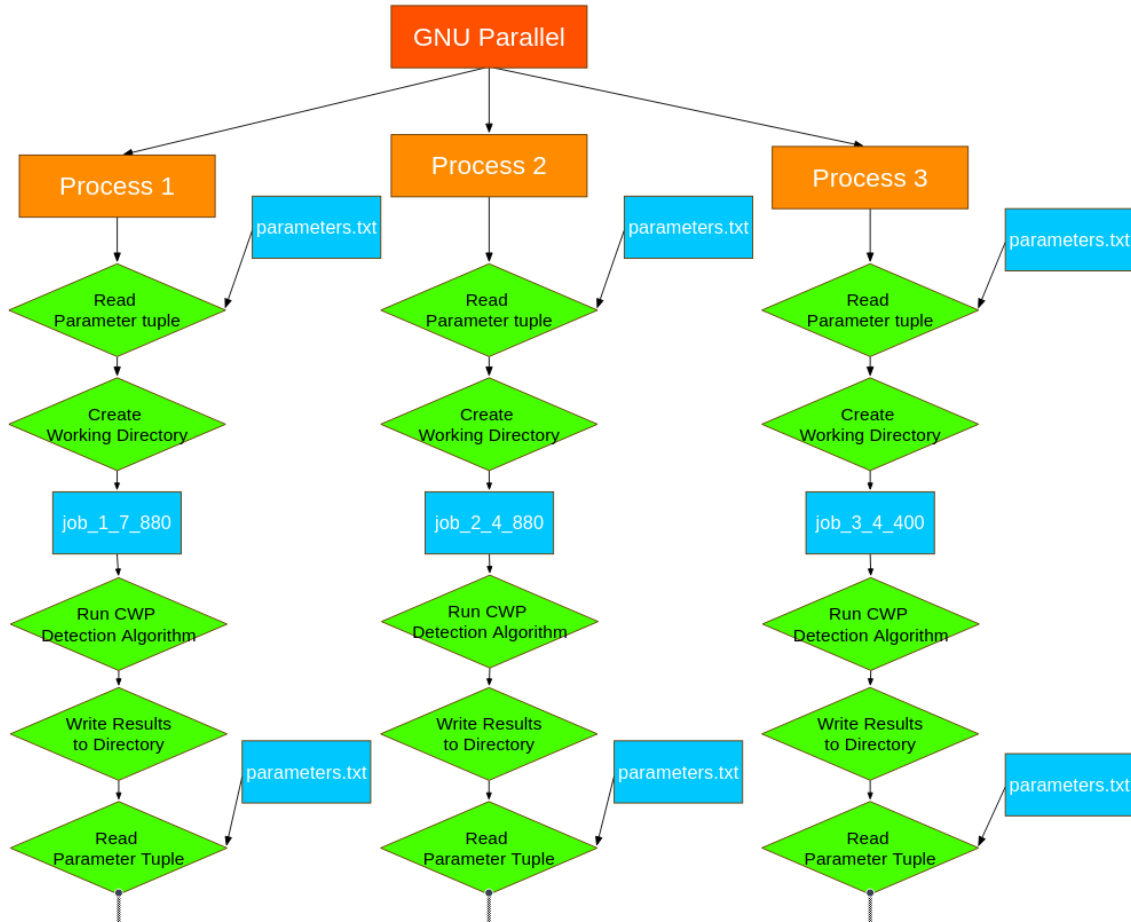


Fig. 3-4: Worker pool orchestrated by GNU Parallel across three parallel processes. Each process reads a parameter tuple from a shared `parameters.txt` file, creates an individual working directory named after the tuple identifier and parameter values (e.g. `job_1_7_880`), executes the CWP detection algorithm, and writes the results to that directory before fetching the next parameter tuple. The cycle repeats until all tuples have been processed.

Creation of a Fuzzy Overlay of CWP

The CWP polygon layers from all runs were rasterized at the TIR raster resolution, stacked into a multi-layer raster, and reduced to a single two-dimensional raster via pixel-wise mean. The result is visualized in Figure 3-5.

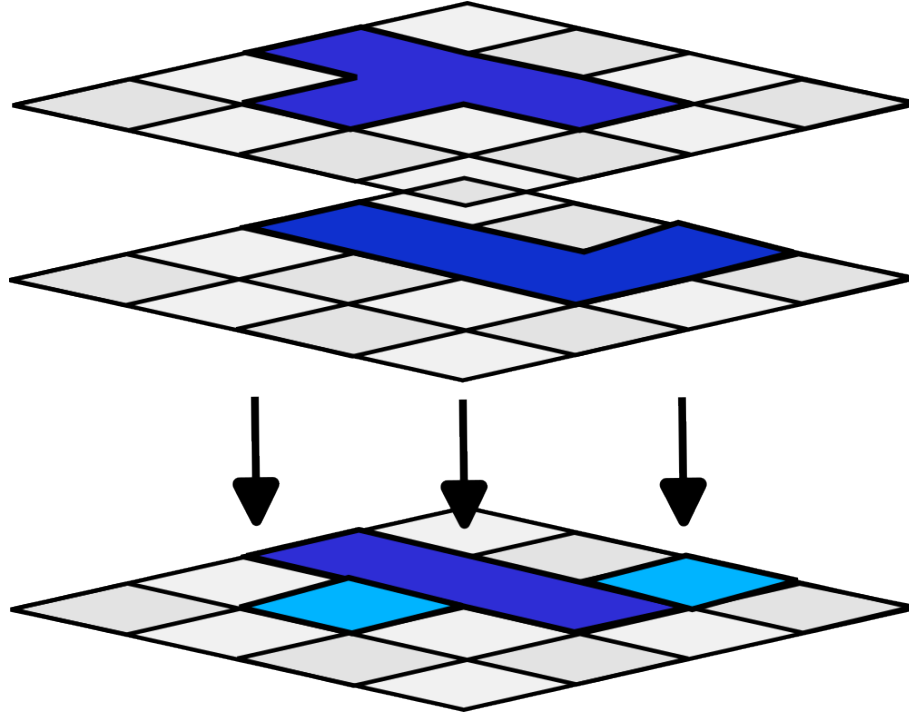


Fig. 3-5: Illustration of the fuzzy overlay construction. The two upper rasters show the binary CWP detection outputs of two individual Morris runs, where dark blue pixels indicate detected CWP and white pixels indicate non-CWP area. The lower raster shows the resulting fuzzy overlay, where light blue pixels hold a lower score than dark blue pixels, reflecting locations where CWP detection was inconsistent across runs.

Each pixel in the resulting raster holds a score between 0 and 1, representing the fraction of runs in which that pixel was detected as part of a CWP – interpretable as a fuzzy membership score. CWPs showing a wide range of scores across their extent are variable with respect to the step length parameter and are therefore candidates for the selection and classification process described in the following subsection.

Selection and Classification of CWP

From the fuzzy CWP raster, patches that were clearly variable and clearly attributable to either a tributary inflow or its absence were selected and classified as tributary-caused or non-tributary-caused based on visual interpretation. Variability was assessed from the fuzzy overlay coloring, and tributary attribution was determined by proximity to a tributary inflow. Figure 3-6 visualizes this as a decision tree.

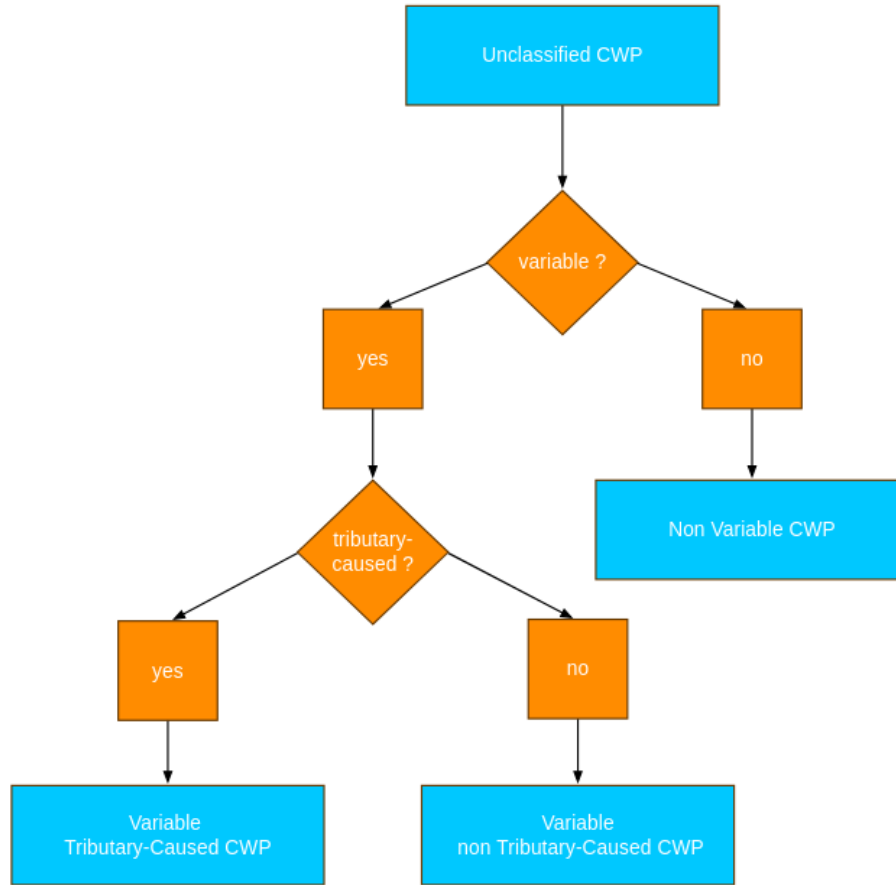


Fig. 3-6: Decision tree illustrating the visual classification process applied to candidate CWP patches: patches are first assessed as variable or non-variable based on their coloring in the fuzzy overlay, and variable patches are further classified as tributary-caused or non-tributary-caused based on their proximity to a tributary inflow.

Tributary inflows were identified using OpenStreetMap and SwissImage in combination, as neither source can be assumed complete on its own. A total of 24 CWPs were selected and classified; for each, a bounding box was extracted to serve as the spatial unit for the sensitivity index computation.

Multiple small adjacent CWPs were grouped and treated as a single patch. This decision is motivated by Tobler's first law of geography, which states that all things are related, but close things are more related than distant things. CWPs in close spatial proximity share the same local generating conditions – the same tributary, the same reach geometry, the same local thermal regime – and will therefore exhibit highly similar sensitivity to the step length parameter precisely because of that proximity. Treating them as separate observations would consequently over-represent that location in the analysis, violating the assumption of independence between observations and skewing the sensitivity indices towards the conditions prevailing at that particular spot.



(a) Variable, tributary-caused. Coloring indicates variability, and the patch lies adjacent to a tributary inflow.



(b) Variable, non-tributary-caused. Coloring indicates variability, but no tributary inflow is present nearby.



(c) Non-variable, tributary-caused. The patch shows little variability in the fuzzy overlay despite proximity to a tributary inflow.



(d) Non-variable, non-tributary-caused. The patch shows little variability and no tributary inflow is present nearby.

Fig. 3-7: Examples of the four classification outcomes from the decision tree in Figure 3-6: variable vs. non-variable patches (rows), further split by tributary attribution (columns).

4.3.2 Computing Morris Sensitivity Indices with respect to CWP Area

With the 24 CWPs classified, the second stage quantifies their sensitivity to buffer pixel count, step length, and temperature delta. Following [Lilburne and Tarantola \(2009\)](#), the spatially distributed polygon output is aggregated to a scalar statistic per run – here, the total detected CWP area within each bounding box. Figure 3-8 summarises the steps involved.

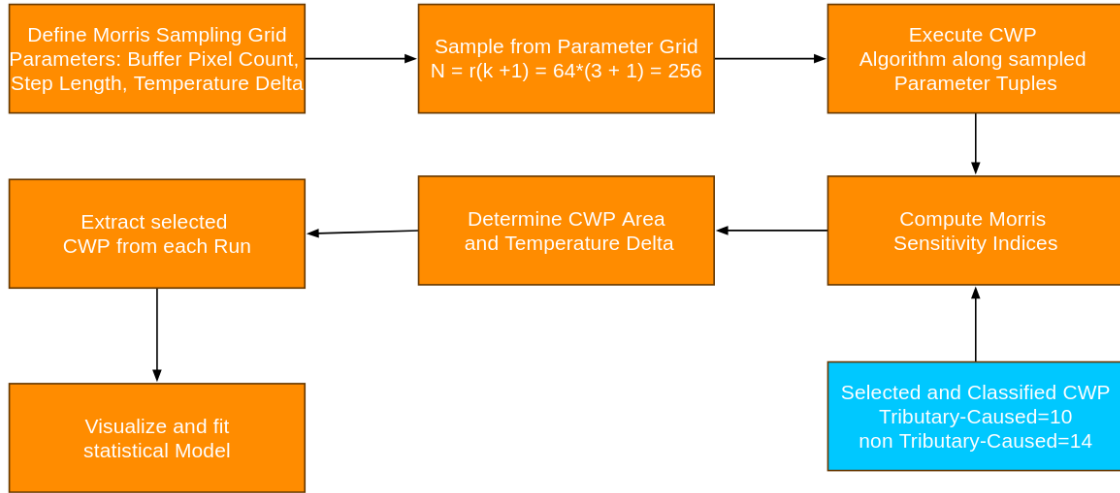


Fig. 3-8: Visual overview of the steps involved in computing Morris sensitivity indices with respect to CWP area: setting up the second Morris experiment ($k = 3$) and executing the algorithm across the sampled parameter tuples, computing lumped area statistics per annotated CWP location and Morris run, and computing and visualising the Morris sensitivity indices.

Setting up Second Sampling Grid and Executing Algorithm

A three-dimensional Morris sampling grid spanning buffer pixel count, step length, and temperature delta was set up, as depicted in Figure 3-9. A total of 64 trajectories were sampled ($r = 64$), yielding $64 \times (3 + 1) = 256$ parameter tuples (Equation 2-3). The algorithm was executed in parallel using the same orchestration as in Section 4.3.1, with the remaining parameters held constant as listed in Table 3-5.

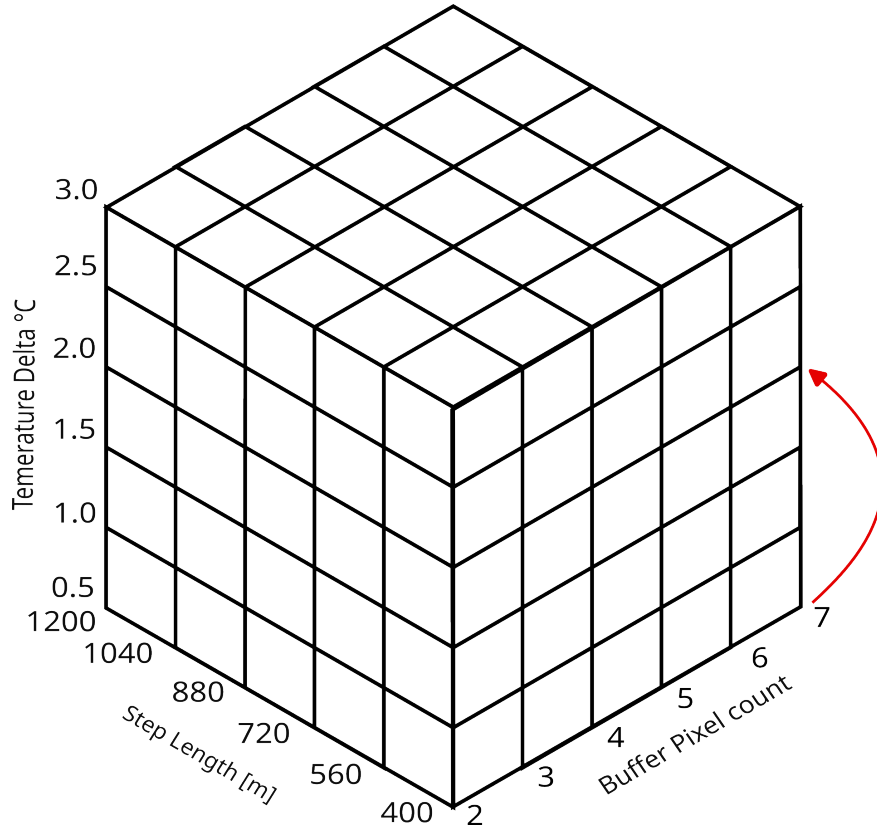


Fig. 3-9: Visual depiction of the second Morris sampling grid, spanning step length, buffer pixel count, and the temperature delta threshold. The red arrow indicates the Δ chosen.

Tab. 3-5: Parameter values held constant during the second Morris experiment, while `step_length`, `buffer_px`, and `delta_T` were varied.

Parameter	Value
<code>min_patch_area_m2</code>	2 m ²
<code>zone_halfwidth_m</code>	60 m
<code>connect_diagonals</code>	TRUE

Computing Lumped Statistics for the Selected CWPs

For each of the 24 annotated bounding boxes and each Morris run, the CWP polygons inside the bounding box are extracted and the total detected CWP area and the area-weighted temperature delta are computed. If no CWP is detected for a given run, both statistics are set to zero. Figure 3-10 illustrates this computation.

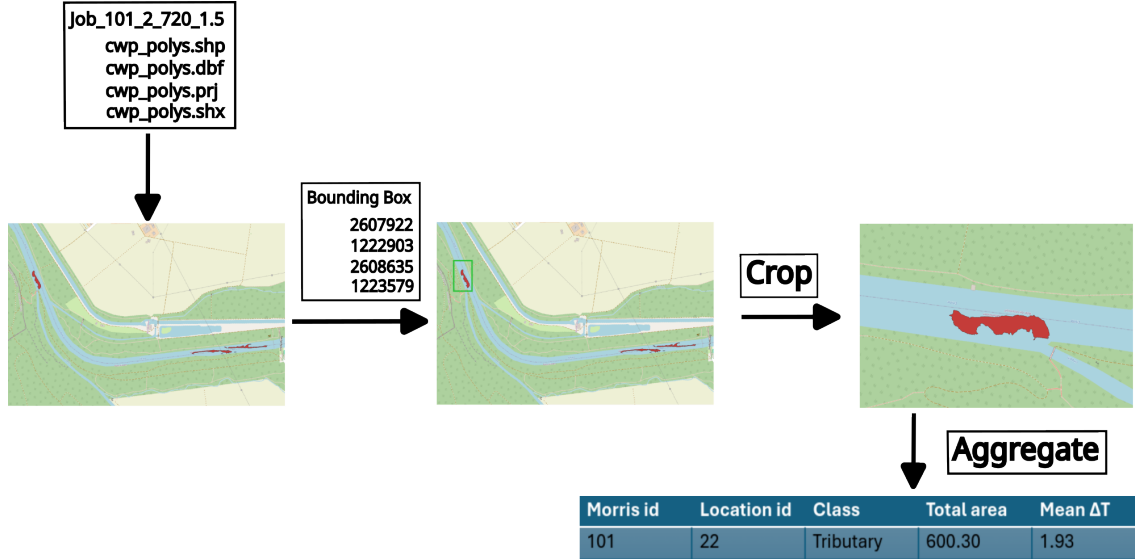


Fig. 3-10: Schematic overview of the lumped-statistics computation for one annotated CWP location (Limpbach inflow into the Emme near Utzenstorf BE). For each Morris run, CWP polygons are cropped to the bounding box; if polygons are present, total area and area-weighted mean temperature delta are computed; if the bounding box is empty, both are set to zero.

The result is a table in which each row corresponds to one annotated location under one parameter tuple. Table 3-6 shows an excerpt; the varying area values across rows sharing the same location confirm that total CWP area is a suitable scalar statistic for characterising parameter sensitivity.

Tab. 3-6: Excerpt of the lumped statistics table. Each row corresponds to one annotated CWP location under one Morris parameter tuple.

Location	Class	ΔT [°C]	Total CWP area [m ²]
2	Non-tributary	3.67	32.32
2	Non-tributary	4.13	22.65
2	Non-tributary	4.28	18.92
⋮	⋮	⋮	⋮
4	Tributary	2.40	39.82
4	Tributary	2.91	15.89
4	Tributary	0	0
⋮	⋮	⋮	⋮

Computing Morris Sensitivity Indices

Before computing the sensitivity indices, the total CWP area values are normalised per location using min-max scaling into $[0, 1]$. This ensures comparability across locations: without normalisation, locations with inherently larger CWP would produce larger absolute area differences, inflating their sensitivity indices relative to smaller patches. After scaling, a value of 1 corresponds to the maximum area detected for that location across all runs and a value of 0 to an empty bounding box.

The Morris indices μ , μ^* , and σ are then computed with respect to the normalised area values using the R package `sensitivity`.

Visualisation and Statistical Modelling

Two sets of boxplot visualisations were produced. The first displays μ^* and σ with respect to step length, stratified by CWP class, to allow a direct comparison of sensitivity between tributary-caused and non-tributary-caused CWPs. The second displays μ^* for all three varied parameters pooled across both classes, providing an overview of their relative influence on CWP area.

Additionally, a linear model was fitted to assess whether CWP area and temperature delta are associated with step-length sensitivity:

$$\mu_{\text{step length}}^* = \beta_0 + \beta_1 \cdot \overline{\text{area}} + \beta_2 \cdot \overline{\Delta T} + \varepsilon \quad (3-3)$$

where $\mu_{\text{step length}}^*$ is the Morris sensitivity index for step length, $\overline{\text{area}}$ and $\overline{\Delta T}$ are the mean total CWP area and mean temperature delta across all Morris runs for each annotated location, and ε is the residual error. The model was fitted using R's `lm()` across all 24 locations, pooling both classes.

4.4 Addressing Research Question 3: Algorithm Modification

Building on the sensitivity analysis findings of [Tonolla and Antonetti \(2026\)](#), this section describes the design and implementation of a modified algorithm, referred to as v3, that addresses the identified parameter sensitivities. The section covers the motivation for each design decision, the operationalisation of the revised reference temperature computation, the implementation details, and the resulting reduction in free parameters.

4.4.1 Motivation for Algorithm Modification

The sensitivity analysis of [Tonolla and Antonetti \(2026\)](#) provides direct motivation for the design of v3. Three findings from that work informed specific algorithmic decisions.

First, [Tonolla and Antonetti \(2026\)](#) found that the buffer pixel count has little influence on the detected CWP area. This suggests that the buffer pixel count qualifies as a candidate for factor fixing – that is, it can be set to a fixed constant and removed as a free parameter without meaningfully affecting the algorithm's output. In v3, the buffer pixel count is therefore internalized and set to a constant value of 3, which lies within the range of values examined by [Tonolla and Antonetti \(2026\)](#).

Second, [Tonolla and Antonetti \(2026\)](#) identified the step length parameter as having a non-negligible influence on the detected CWP area. A plausible mechanistic explanation for this sensitivity is that CWPs spanning a slab boundary are compared against two different reference temperatures – one per slab – which can produce artificial truncations in the detected patch extent. This motivated a fundamental redesign of the reference temperature computation in v3, replacing the slab-based approach with a spatially continuous IDW-based scheme that does not depend on a step length parameter at all.

Third, [Tonolla and Antonetti \(2026\)](#) found the temperature delta parameter to be the most influential of the parameters examined. Unlike the buffer pixel count and step length, however, the temperature delta cannot simply be removed: different thresholds are used in the literature to define CWPs ([Casas-Mulet et al., 2020](#)), and the user must therefore retain the ability to set it according to their chosen definition. The temperature delta is accordingly retained as a free parameter in v3.

Whether the design decisions described above are supported by the Morris screening analysis conducted in this work will be examined in the Results and Discussion.

4.4.2 Operationalisation of Surrounding Stream Temperature

The step-length dependency was addressed by rethinking the operationalisation of the surrounding stream temperature used as the reference for pixel flagging. In v1 and v2, the reference temperature for each pixel is determined by the median temperature of the slab segment to which that pixel belongs. This approach has two consequences: it requires a user-defined step length, and it produces a reference temperature field that is piecewise constant – uniform within each slab and discontinuous at slab boundaries. CWPs that happen to span a slab boundary are therefore compared against two different reference temperatures, which can produce artificial cutoffs in the detected patch extent.

In v3, this is replaced by an inverse distance weighting (IDW) scheme. Rather than dividing the reach into fixed slabs, reference temperature sample points are obtained by sampling multiple step length values from a uniform distribution and extracting the median temperature at each resulting segment’s visual center. These sample points, pooled across all sampled step lengths, are then used to compute a spatially continuous reference temperature raster via IDW, where each cell’s reference temperature is a distance-weighted average of the surrounding sample point temperatures. This guarantees that the reference temperature is a spatially continuous field, and that no sharp cutoffs at slab boundaries can occur.

4.4.3 Implementation Detail

The implementation of v3 follows the structure of v2 closely, differing only in the computation of the reference temperature raster (steps 1–5 below, replacing steps 1–5 of [Table 2-2](#)). The full reference temperature computation proceeds as follows:

Tab. 4-7: High-level overview of the v3 CWP detection algorithm. Steps 1–5 replace the corresponding steps of Table 2-2; steps 6–8 are identical to v2.

Step	Description
1	Sample 20 candidate step length values from a uniform distribution with bounds 500–3000 metres.
2	For each candidate step length, divide the river centerline into n segments; construct a narrow 3-pixel-wide reference strip polygon for each segment; extract the median temperature of each segment from the TIR raster; compute the visual centre of each strip polygon and assign the corresponding median temperature to this point.
3	Pool the reference points (location + temperature) from all 20 step length values into a single combined set of IDW sample points.
4	Initialise a new raster with the same extent and resolution as the TIR raster. For each cell: if the corresponding TIR cell is NaN, set to NaN; otherwise compute a distance-weighted reference temperature via IDW over the pooled sample points, using a decay power of 2.
5	Round the TIR raster using a fixed rounding factor.
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT , producing a binary CWP raster.
7	Polygonize the binary CWP raster into patch polygons.
8	Filter patches by minimum area; compute per-patch temperature statistics; apply a patch-level threshold check against the reference temperature.

Step 4 is implemented in C++ and imported into R via the `Rcpp` package, as the per-pixel IDW computation was identified as the primary computational bottleneck.

4.4.4 Reduction of Parameters

The buffer pixel count is fixed internally at a value of 3. The step length parameter is no longer required, as the IDW reference temperature computation samples reference points across a range of step lengths. The zone halfwidth parameter is likewise no longer needed, as the extent of the reference raster is matched to the extent of valid TIR pixels – though this implies a constraint: v3 requires a TIR raster with an exact water mask, with non-river pixels set to NaN. The remaining free parameters are: the TIR raster and centerline paths, minimum patch area, rounding factor, temperature delta threshold, and diagonal connectivity.

4.5 Usage of AI

The large language models Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic) were used as aids in this project. Detailed insight into the usage is given in Appendix A.

Chapter 5

Results

5.1 Research Question 1: Algorithm Reimplementation

This section presents the results of the runtime benchmarking and output equivalence comparison between v1 and v2. The two subsections address the speed improvement achieved by the reimplementation and the degree to which the two implementations produce spatially consistent CWP detections.

5.1.1 Runtime Comparison

Figure 1-1 shows the execution time of v1 and v2 as a function of segment length. The execution time of v2 is approximately 0.5 hours across all three tested segment lengths, remaining largely constant. In contrast, v1 shows execution times ranging from approximately 2 hours at a segment length of 400 m to approximately 1.25 hours at 1200 m, with execution time increasing as segment length decreases (i.e., as the number of segments increases).

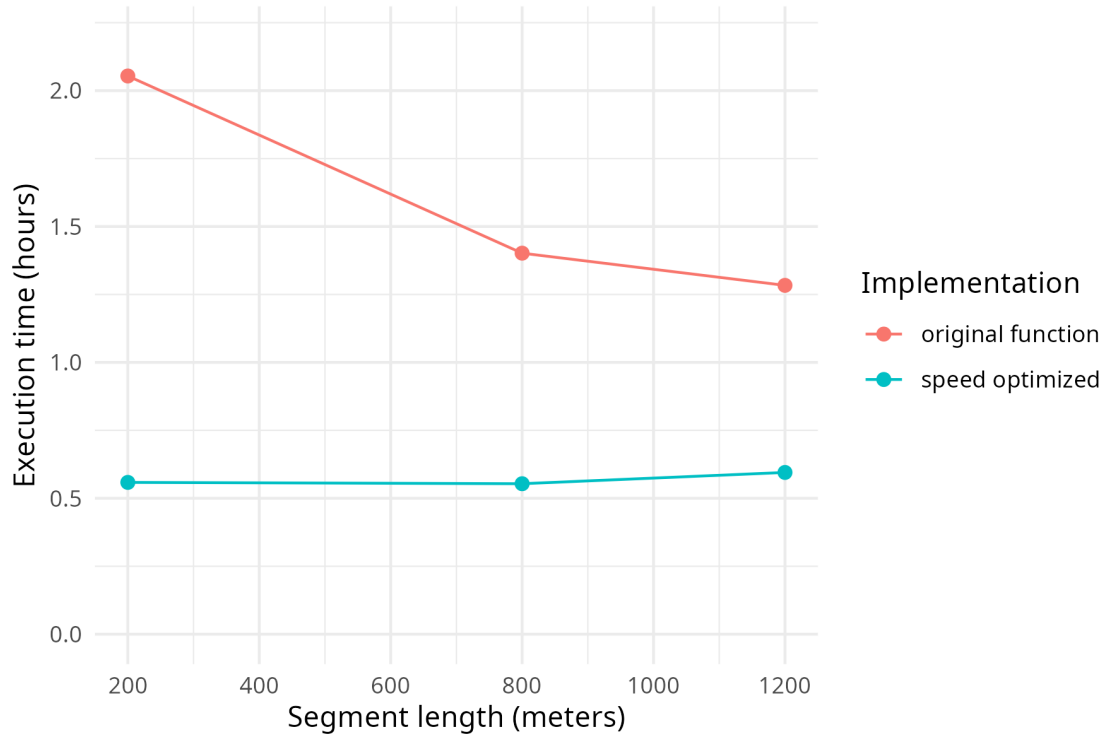
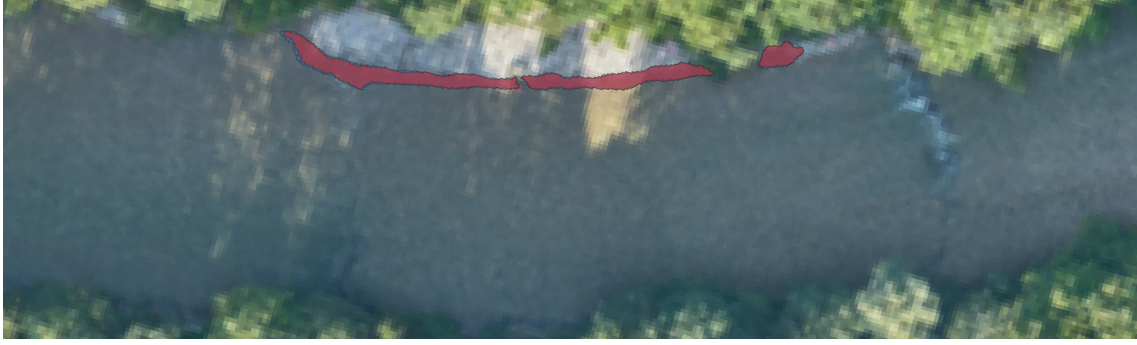


Fig. 1-1: Execution time of the original (v1) and optimized (v2) implementations of `detect_cwp_single` as a function of segment length. The optimized implementation shows execution times that are both shorter and largely constant across segment lengths, whereas the original implementation scales disproportionately with decreasing segment length (i.e., increasing number of segments).

5.1.2 Output Equivalence

The three comparison runs achieved Jaccard similarities of 0.936, 0.956, and 0.976, indicating generally good spatial agreement between v1 and v2. In all panels, v2 polygons are shown in red and v1 polygons in blue, overlaid at 50% transparency. Figure 1-2 illustrates the range of agreement observed. Panel (a) shows a representative example of general agreement, with discrepancies confined to a small number of boundary pixels. Panel (b) shows a more pronounced boundary mismatch. Panel (c) shows an example of complete spatial agreement, where the two polygon sets are visually indistinguishable.



(a) General agreement. Discrepancies are confined to patch boundaries.



(b) More pronounced disagreement. Boundary mismatches between v1 and v2 are more evident.



(c) Complete agreement. v1 and v2 polygons are spatially indistinguishable.

Fig. 1-2: Comparison of CWP polygon outputs between v1 (blue) and v2 (red) across the three validation runs. Polygons are overlaid at 50% transparency; areas of agreement appear as a blend of both colours, while areas covered by only one algorithm appear in that algorithm's colour alone.

5.2 Research Question 2: Sensitivity Analysis

This section presents the Morris sensitivity analysis results across three complementary views: the sensitivity of detected CWP area to the step length parameter stratified by CWP class, the overall sensitivity ranking across all three varied parameters, and the relationship between CWP area and step-length sensitivity as quantified by the linear model.

5.2.1 Sensitivity of CWP Area to Step Length

Figure 2-3 shows the Morris sensitivity indices μ^* and σ with respect to the step length parameter, stratified by CWP class ($n = 10$ tributary-caused, $n = 14$ non-tributary-caused). All non-tributary-caused CWP show μ^* values below 0.2, indicating low sensitivity to the step length parameter across this class. The tributary-caused CWP show a wider spread in μ^* : some tributary-caused CWP have sensitivity values comparable to the non-tributary-caused group, while others show substantially higher values of up to 0.3–0.6. The median μ^* values of the two classes are similar, with the primary difference lying in the upper tail of the tributary-caused distribution. The σ values show a comparable pattern, with greater spread in the tributary-caused group.

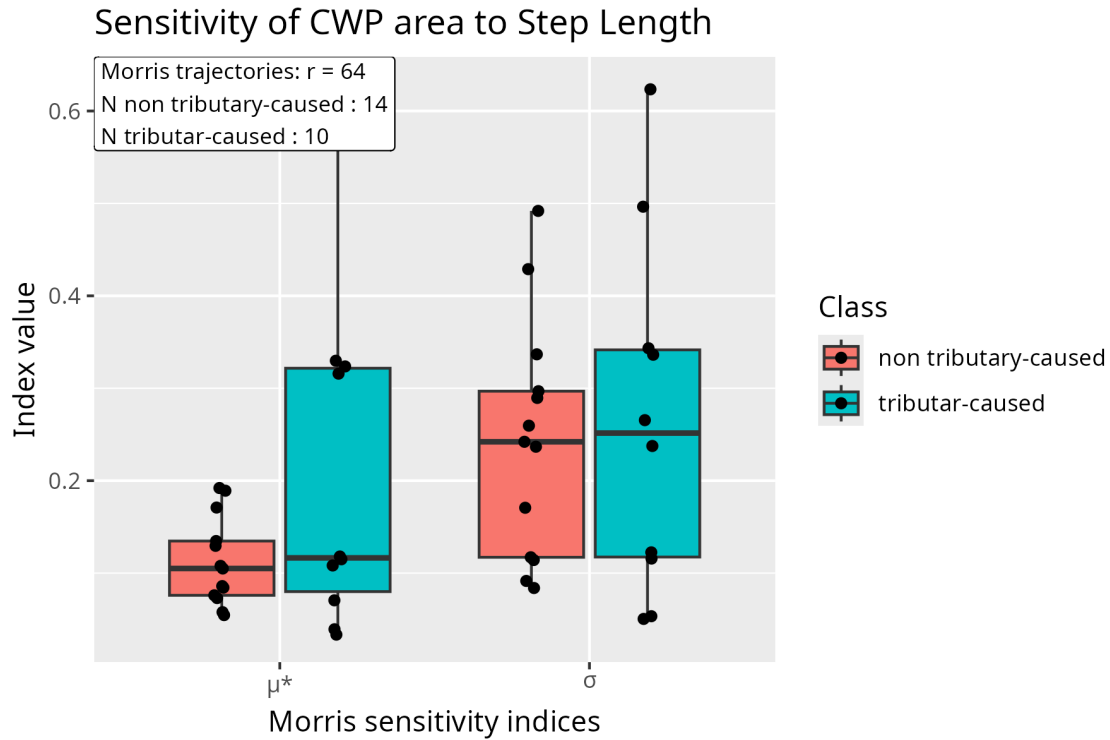


Fig. 2-3: Morris sensitivity indices (μ^* and σ) for the `step_length` parameter with respect to cold water patch area, stratified by patch class (red: tributary-caused, $n = 10$; blue: non-tributary-caused, $n = 14$). Boxplots show the distribution across bounding boxes.

5.2.2 Overall Sensitivity across all Parameters

Figure 2-4 displays the μ^* values for all 24 CWP locations with respect to the three parameters varied in the second Morris experiment. The buffer pixel count shows consistently low μ^* values across all locations, with the entire distribution concentrated near zero. In contrast, both step length and temperature delta show distributions that deviate clearly from zero. The temperature delta parameter shows the highest μ^* values overall, suggesting it has the strongest influence on the detected CWP area across the set of annotated locations.

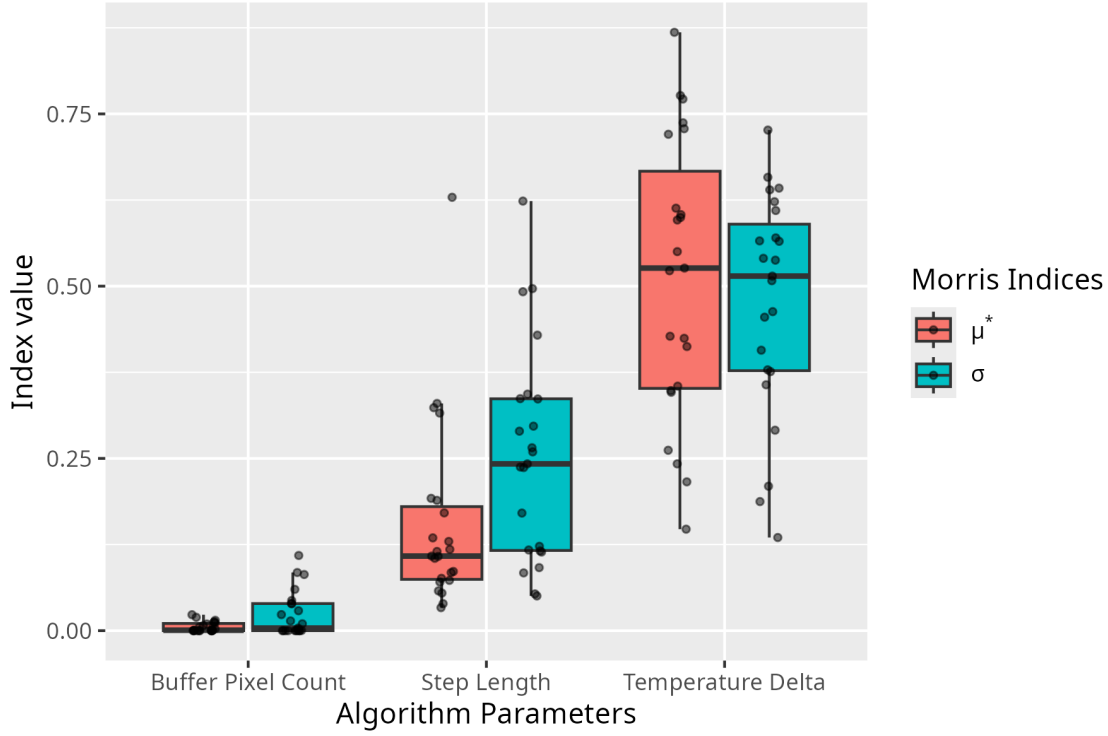


Fig. 2-4: Overall μ^* sensitivity indices for `buffer_px`, `delta_T`, and `step_length` across all classified cold water patches ($N = 24$ locations, $r = 64$ trajectories).

5.2.3 Relationship between CWP Area and Step Length Sensitivity

Figure 2-5 shows the 24 selected CWP along three dimensions simultaneously: mean CWP area averaged across all Morris runs (x-axis), μ^* with respect to the step length parameter (y-axis), mean temperature delta to the surrounding environment averaged across all Morris runs (color), and CWP class (shape). The data suggest a positive relationship between mean CWP area and sensitivity to the step length parameter: locations with larger mean CWP area tend to show higher μ^* values. No clear relationship between mean temperature delta and μ^* is visible from the colouring.

Table 2-1 shows the results of the linear model. The mean area predictor is highly significant ($p < 0.001$) with a positive slope, consistent with the visual pattern in Figure 2-5. The mean temperature delta predictor is not significant ($p = 0.556$). The model explains a substantial proportion of variance ($R^2 = 0.611$, Adjusted $R^2 = 0.572$).

Tab. 2-1: Regression results for μ^* (step length) predicted by mean CWP area and mean ΔT , fitted across all 24 annotated locations.

Predictor	Estimate	SE	t	p
(Intercept)	9.024×10^{-2}	3.094×10^{-2}	2.92	0.009
Mean area	1.578×10^{-4}	2.839×10^{-5}	5.56	< 0.001
Mean ΔT	-1.041×10^{-2}	1.739×10^{-2}	-0.60	0.556

Note. $R^2 = 0.611$, Adjusted $R^2 = 0.572$, $F(2, 20) = 15.73$, $p < 0.001$. Residual standard error = 0.088 on 20 degrees of freedom (one observation deleted due to missingness).

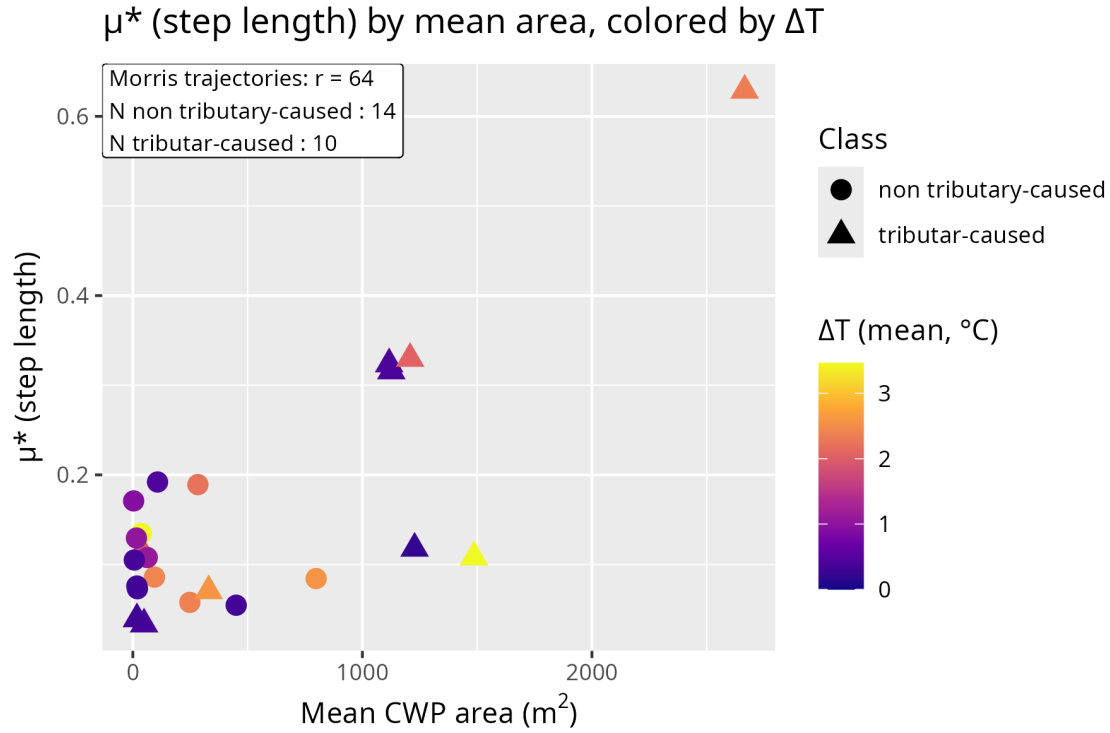


Fig. 2-5: μ^* (step length) versus mean CWP area for each of the 24 annotated locations. Color indicates the mean temperature delta to the surrounding river ($\overline{\Delta T}$, averaged across all Morris runs); shape indicates CWP class (circle: non-tributary-caused; triangle: tributary-caused).

5.3 Research Question 3: Algorithm Modification

This section presents the evaluation of v3 relative to v1, covering both the quantitative Jaccard similarity comparison and the qualitative assessment of detection improvements resulting from the IDW-based reference temperature computation.

5.3.1 Output Comparison of v3 against v1

The CWP polygons produced by v3 were compared against those of v1 using the same Jaccard similarity metric applied in the output equivalence comparison of v1 and v2 (Section 4.2.2). The Jaccard similarities obtained were 0.325, 0.334, and 0.287 for the three validation parameter combinations – substantially lower than the values of 0.936, 0.956, and 0.976 obtained for v2 versus v1. This indicates that v3 produces CWP detections that differ considerably from those of v1, which is expected: the IDW-based reference temperature computation represents a fundamental redesign of the core reference temperature step rather than a minor reimplementation, and is therefore not expected to replicate v1's outputs closely. The relevant question is not whether v3 agrees with v1, but whether v3's detections are ecologically more plausible.

Figure 3-6 illustrates three qualitative improvements observed in v3 relative to v1. In each panel, v1 polygons and v3 polygons are overlaid with 50% transparency, allowing areas of agreement and disagreement to be read directly from the overlap pattern.



(a) Slab-boundary truncation eliminated. The v1 polygon ends abruptly at the slab boundary; the v3 polygon extends naturally beyond it.



(b) Merger of previously severed fragments. Two separate v1 polygons are detected as a single contiguous patch in v3.



(c) New detection in v3. A CWP absent in v1 is detected by v3, consistent with the hypothesis that it was split into sub-threshold fragments by a slab boundary in v1.

Fig. 3-6: Qualitative comparison of CWP detections between v1 and v3 at three example locations. In each panel, v1 polygons and v3 polygons are overlaid at 50% transparency; areas covered by both algorithms appear as a blend of the two colours, while areas covered by only one algorithm appear in that algorithm's colour alone.

Panel (a) shows the most direct improvement: in v1, the detected CWP extent is truncated abruptly at a slab boundary, not because the underlying thermal anomaly ends there, but because the reference temperature changes discontinuously at that point. In v3, the spatially continuous IDW reference temperature surface removes this artificial boundary, allowing the detected patch to extend naturally beyond the former boundary location and taper off gradually as the temperature difference to the surrounding river diminishes.

Panel (b) illustrates a direct consequence of eliminating slab-boundary truncations: CWPs previously reported as two separate fragments – one on each side of a slab boundary – are

now detected as a single contiguous polygon, which is likely a more accurate representation of the underlying thermal anomaly.

Panel (c) shows an example of a CWP detected by v3 that is entirely absent in v1. A likely explanation is that the corresponding thermal anomaly straddles a slab boundary in v1, causing it to be split into two sub-threshold fragments that are subsequently removed by the minimum area filter. In v3, the absence of slab boundaries allows the anomaly to be detected as a single contiguous patch above the area threshold. Whether these additional detections represent genuine CWPs missed by v1, or false positives introduced by the IDW reference temperature surface, cannot be determined without independent ground-truth validation – but the former interpretation is consistent with the mechanistic argument developed in Section 4.4.

Chapter 6

Discussion

6.1 Research Question 1: Algorithm Reimplementation

This section discusses the results of the output equivalence comparison and the runtime benchmarking between v1 and v2, and assesses whether the sensitivity analysis findings derived from v2 can be transferred to v1.

6.1.1 Difference in CWP Polygons between v1 and v2

The CWP polygons produced by v1 and v2 are in general agreement with respect to location and spatial extent, as reflected by Jaccard similarities of 0.936, 0.956, and 0.976 across the three validation runs. The observed discrepancies are confined to the pixel level – typically affecting a small number of pixels at patch boundaries – and no systematic spatial bias is apparent.

The most likely source of these discrepancies is the difference in how `terra::extract()` and `exactextractr::exact_extract()` handle partially covered pixels: the 50% coverage-fraction threshold used to approximate `terra`’s centre-point inclusion rule is not exactly equivalent, and small differences in extracted reference temperatures propagate to pixel-level differences in the binary CWP raster.

The high Jaccard similarities, combined with the observation that the discrepancies can be attributed to a minor extraction artefact rather than a fundamental algorithmic difference, support the conclusion that v2 replicates the behaviour of v1 to a degree sufficient to transfer sensitivity findings.

6.1.2 Speed Comparison

The optimized algorithm (v2) shows substantially shorter execution times than v1 across all three tested segment lengths, with the performance advantage most pronounced at shorter step lengths. While v2 maintains an approximately constant execution time of around 0.5 hours regardless of step length, v1’s execution time increases as step length decreases, ranging from approximately 1.25 hours at 1200 m to approximately 2 hours at 400 m.

This difference in scaling behaviour is consistent with a mechanistic explanation rooted

in how the two implementations process the river reach. v2 is loop-free: GDAL operates on the full raster in a single pass, and each pixel undergoes a fixed number of operations regardless of how many segments the river reach is divided into. v1, by contrast, processes the reach one slab at a time using R loops. Each loop iteration incurs a fixed overhead independent of slab size, plus a variable computation proportional to the raster area of that slab. At shorter step lengths the reach is divided into more, smaller slabs, shifting the ratio of overhead-to-useful-work unfavourably and increasing total execution time. It should be noted that this mechanistic account is a hypothesis and was not formally tested within this work.

v2 is faster than v1 across the entire parameter range tested, and the performance advantage is largest at the shorter step lengths (400 m) where v1 is slowest – which fall within the Morris sampling grid used in this work.

6.2 Research Question 2: Sensitivity Analysis

This section interprets the Morris sensitivity analysis results, first addressing the specific hypothesis regarding step-length sensitivity in tributary-caused versus non-tributary-caused CWP, and then discussing the overall parameter sensitivity ranking across all three varied parameters.

6.2.1 Answering Research Question 2

The boxplot comparison of μ^* values between tributary-caused and non-tributary-caused CWP shows a tendency consistent with H1: tributary-caused CWP display a wider spread in step-length sensitivity, with a higher third quartile and higher maximum values than the non-tributary-caused group. However, a closer inspection reveals that this pattern is not uniform within the tributary-caused class – some tributary-caused CWP show sensitivity values comparable to the non-tributary-caused group, while others are substantially more sensitive. This within-class heterogeneity suggests that tributary presence alone is not a sufficient condition for high sensitivity to the step length parameter, and that H1 as stated does not fully capture the situation. At the same time, the data do not support H0 either: the non-tributary-caused CWP consistently show low μ^* values, and the elevated sensitivity observed in a subset of tributary-caused CWP is unlikely to be entirely coincidental. The true picture lies between the two hypotheses.

The scatter plot and linear model together point to CWP area as the more proximate driver of step-length sensitivity. Across both classes, locations with larger mean CWP area tend to show higher μ^* values, and this relationship is approximately linear. The linear model confirms this: the mean area predictor is highly significant ($p < 0.001$) with a positive slope, while the mean temperature delta predictor is not significant ($p = 0.556$).

A mechanistic explanation for the area-sensitivity relationship follows naturally from the structure of the algorithm. CWPs in a river system tend to be elongated rather than wide, since the river channel constrains lateral extent while longitudinal extent is less bounded. Large CWP are therefore predominantly long CWP. A long CWP has a higher probability of spanning the boundary between two adjacent slabs, and when this occurs, the two parts of the patch are compared against different reference temperatures – one computed from each slab’s reference strip. If those reference temperatures differ (which is

more likely near a tributary inflow, where sharp longitudinal temperature gradients exist), the flagging threshold is effectively applied inconsistently across the patch, introducing variability in the detected CWP extent that is directly attributable to the step length parameter. Shorter or smaller CWP are more likely to fall entirely within a single slab, and are therefore less exposed to this effect.

Tributary inflows are then best understood not as a direct cause of step-length sensitivity, but as one mechanism that can produce large, elongated CWP – by introducing substantial volumes of cold water that generate extended thermal anomalies along the river. It also means that the data cannot speak to whether a large non-tributary-caused CWP would show comparable sensitivity to an equally large tributary-caused CWP, since no non-tributary-caused CWP larger than approximately 1000 m² is present in the dataset.

Interestingly, CWP area and mean temperature delta appear to be largely unrelated in the data, despite the intuitive expectation that a colder tributary might produce a larger thermal anomaly. This could suggest that tributary discharge plays a more important role than temperature difference in determining CWP size – a large, moderately cold tributary may produce a larger CWP than a small but very cold one. Discharge data were not collected during the survey, but could be a valuable addition in future work.

Neither H0 nor H1 is fully supported by the data. H0 is too strong a rejection – tributary presence is associated with a tendency towards higher step-length sensitivity. However, H1 overstates the role of tributary presence as a direct determinant of sensitivity. The data suggest that the more proximate driver is CWP size: large CWP, which are disproportionately but not exclusively tributary-caused, are more susceptible to the step length parameter because they are more likely to span slab boundaries and therefore be exposed to inconsistent reference temperatures.

6.2.2 Overall Sensitivity across all Parameters

The overall μ^* distributions across all 24 annotated locations reveal a clear ranking of parameter influence with respect to detected CWP area. The buffer pixel count is consistently non-influential, with μ^* values close to zero across all locations. The corresponding σ values are similarly low, suggesting that the elementary effects of buffer pixel count are both small in magnitude and consistent across the parameter space – indicating neither a strong main effect nor meaningful interactions with the other parameters. This finding is consistent with [Tonolla and Antonetti \(2026\)](#), who report the same result in their sensitivity analysis of the CWP detection algorithm. Given this replication, the strategy of factor fixing can be applied: the buffer pixel count can be set to a fixed constant value, effectively removing it as a free parameter of the algorithm.

The temperature delta parameter shows the highest μ^* values overall, indicating that it is the most influential of the three parameters with respect to CWP area. This is also consistent with the findings of [Tonolla and Antonetti \(2026\)](#), who likewise identify temperature delta as the dominant parameter. The elevated σ values for temperature delta suggest that its elementary effects vary across the parameter space, pointing to the presence of nonlinearity or interactions with the other parameters. This is ecologically plausible: the relationship between the flagging threshold and the detected CWP area is unlikely to be linear, as small changes in temperature delta near the boundary of what constitutes a detectable thermal anomaly could produce disproportionately large changes in detected area, while changes far from this boundary may have little effect.

The step length parameter occupies an intermediate position, with μ^* values clearly above zero but consistently below those of temperature delta. Like temperature delta, step length also shows elevated σ values, suggesting interactions with the other parameters or nonlinear behaviour.

6.3 Research Question 3: Algorithm Modification

The replication of the buffer pixel and temperature delta findings from [Tonolla and Antonetti \(2026\)](#) within the present work, using an independently implemented Morris experiment on a different dataset, lends additional credibility to both sets of results and suggests that these sensitivity patterns are robust properties of the CWP detection algorithm rather than artefacts of a particular study context. This agreement between the two analyses provides a stronger basis for the design decisions underlying v3 than either analysis alone would afford.

Based on these findings, three modifications were implemented in v3, as described in [Section 4.4](#): the buffer pixel count was fixed by factor fixing, the slab-based reference temperature computation was replaced by an IDW scheme removing the step length dependency, and the temperature delta parameter was retained as a user-defined input. The evaluation presented in [Section 5.3](#) shows that v3 produces outputs that differ substantially from v1 in terms of Jaccard similarity, which is expected given the fundamental redesign of the reference temperature step. Qualitatively, however, v3 eliminates the slab-boundary cut-off artefacts identified as a direct consequence of step-length sensitivity, producing more contiguous and naturally extending CWP detections. Research question 3 can therefore be answered affirmatively: the identified parameter sensitivities can be addressed through targeted algorithmic modifications without sacrificing output plausibility.

6.4 Future Improvements

The following subsections identify three methodological limitations of the present work and outline directions in which future studies could address them.

6.4.1 Subjective Selection and Classification of CWP

Both the selection and classification of the CWP were carried out subjectively based on visual inspection of the fuzzy overlay raster. This approach is prone to confirmation bias on two levels. First, the judgment about variability and the judgment about class membership were both made with the study hypothesis already in mind. CWPs that were selected are therefore likely those that were most easily attributable to one of the two classes – patches that clearly appeared variable and clearly appeared tributary-caused or non-tributary-caused. More nuanced cases, where variability was ambiguous or class attribution uncertain, were left out. The resulting dataset is consequently biased towards extreme and obvious cases, and may not be representative of the broader population of CWPs in the Emme system. Second, given that H1 of research question 2 was considered plausible prior to the analysis based on prior observations, the selection process may have been further skewed towards patches that would support H1, even if this influence was not conscious.

Two methodological improvements could reduce these limitations in future work. First, the variability assessment and CWP selection could be replaced by a fully algorithmic approach, removing human judgment from the process. Caution is however warranted: upon defining the rules of a process-based algorithm, the analyst may inadvertently encode the same biases that affect manual selection. Similarly, a data-driven approach would need to learn from a ground truth dataset that is itself produced by human labeling, meaning the bias present in the labeling process would be inherited by the model. Second, if manual classification cannot be avoided, inter-rater reliability assessment should be included: multiple independent observers, each blind to the study hypothesis, could classify the same set of candidate CWPs, and the level of agreement between raters – quantified for example using Cohen’s κ – would provide an empirical measure of classification reliability. Such a dataset could further serve as ground truth for training a data-driven classification algorithm, combining the transparency of manual labeling with the scalability of an automated approach.

6.4.2 Sample Size

With only $n = 24$ annotated CWP locations, the dataset is small by the standards of statistical inference. While the significant area–sensitivity relationship detected by the linear model suggests that the effect is strong enough to be visible at this sample size, the risk of chance findings cannot be excluded. A larger sample would increase statistical power and reduce the probability that the observed relationships are artefacts of the specific locations selected.

6.4.3 Hidden Parameter in v3

Upon closer inspection, v3 introduces a parameter that was not explicitly treated as a free parameter during the development of the algorithm: the decay power of the inverse distance weighting scheme, which governs how quickly the influence of a reference point diminishes with distance. In the current implementation this value is fixed at 2, which is a common default, but the choice is arbitrary and its influence on the detected CWP area has not been assessed. Strictly speaking, this means that the parameter reduction achieved by v3 is incomplete: the decay power has effectively been fixed by convention rather than by a principled sensitivity analysis. A dedicated sensitivity analysis of the IDW decay power would be needed to either justify fixing it at its current value through factor fixing, or to expose it as a free parameter that the user should be able to configure.

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Chapter 7

Appendix

- Appendix A: Catalog of classified CWP
- Appendix B: Statement of Reproducibility and Collaboration
- Appendix C: List of Generative AI Systems

Appendix C: List of Generative AI Systems

The following generative AI systems and AI-assisted tools were utilized in the context of this work:

System Tool	/	Provider & Version	Purpose of Use
Claude Sonnet 4.6		Anthropic (2026) Version Sonnet 4.6	• Stylistic improvement of text • Spelling correction • Code debugging and coding assistance (no agentic coding) • L^AT_EX assistant (debugging and formatting) • Finding relevant literature
Claude Opus 4.6		Anthropic (2026) Version Opus 4.6	• Complex debugging tasks • Complex writing assistance tasks such as assuring cross-chapter naming consistency

Note: The author assumes full responsibility for all generated content, its accuracy, and compliance with the standards of scientific integrity.